This is fictional examination staging, not a historical quotation or a mathematical source. The quiz, learner guide, instructor key, and manuscript govern the finite definitions. The bracket-display restriction below is a chalkboard convention; the paper uses JSON teaching packets.

00: The Staging, the Professor, and the Tribunal of Paper 000

AuthorityEffect: None.
Location: Moscow, Sadovaya 302-bis, Apartment 50.
Presiding Tribunal: Messire Woland, Lev Davidovich Landau ("Dau"), Koroviev-Fagotto, Behemoth, Azazello, Abadonna, Hella; with the Russian-American Delegation: Jeremy, Liudmila, Polina, Yaroslava.
Subject of Examination: Paper 000 — *The Symbol Game: A First Game with $S(n)$, MSS-N, and Basic Packets.*
Companion Documents: - Study Guide: study_guide.md - Examination Paper: quiz.md - Instructor Key: instructor_key.md

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1. The Knock at the Door

The cold autumn sleet over Sadovaya Street had turned into driving grey needles. Water gurgled in the rusted drainpipes of Apartment 50. Inside, the heavy crimson velvet drapes were bolted shut against the Moscow night. A fire roared in the tiled stove, smelling of birch bark, burnt kerosene, and sulfur.

You stood in the dimly lit hallway, soaked through your overcoat, clutching a leather folder filled with notes on smooth manifolds, continuous Lie groups, and elaborate multi-page JSON-RPC envelopes you had downloaded from contemporary AI preprints. In your notebook, you had carelessly scribbled:

"MSS-2 is obviously identical to $S(2)$"

You knocked twice with trembling knuckles.

The heavy oak door swung open without a sound.

Standing on the threshold was **Hella**. Her pale green eyes were cold and unblinking. Across her slender throat ran the thin, vivid crimson scar of the resurrected. She looked down at your hands, saw the leather folder stuffed with JSON envelopes and coordinate charts, and curled her upper lip with disgust:

> *"Leave that paperwork in the umbrella stand. For this chalkboard round, use the bracket display. Keep your JSON packets beside the*

paper; they remain the companion format. If you mistake an envelope for its contents, I will confiscate your chalk."

She stepped aside and pointed a long, bloodless finger toward the candlelit drawing room.

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2. Woland Introduces Dear Professor Landau

In the center of the drawing room sat **Messire Woland**, sunk into a high-backed walnut armchair. His right leg was stretched out onto an embroidered velvet footstool, wrapped in thick wool bandages at the knee. Resting between his knees was his heavy black walking cane, capped with the cast-silver head of a snarling poodle. One eye was black and bottomless; the other sparkled with a piercing, mocking green fire.

"Look at him, Dau," Woland said, his deep bass voice vibrating through the crystal drops of the chandelier. *"Another ambitious traveler from the West. He wishes to play the Symbol Game. He believes that repeating words like 'quantum consciousness', 'differentiable manifold', and 'generative transformer' constitutes a license to create physical law."*

Woland tapped his silver cane against the floor.

"Step out, Lev Davidovich. Let us see if our guest can distinguish two marks on a card without losing his mind."

From the dark alcove near the grandfather clock stepped **Lev Davidovich Landau**—"Dau." Slight, narrow-shouldered, with rebellious black hair and intense, fiery dark eyes, Landau wore an unironed tweed jacket with chalk-dusted lapels. In his fingers, he spun a crisp cylindrical stick of railway chalk with the speed of an electric drill.

Landau stopped the chalk dead in his palm, leveled it at your face, and let out a sharp, dry snort:

> *"Well? What are you staring at? Have you come to take the Theoretical Minimum, or have you come to recite prose from a marketing brochure?
>
> Let me tell you plainly: I do not care about your university degrees, I do not care about your multi-billion-parameter language models, and I do not care about your six-layer JSON wrappers. In my seminar in Kharkiv, if a student confused a coordinate representation with an invariant tensor, I threw him out of the room. Here, Paper 000 asks the simplest question imaginable: you have two marks, a and b. Can you state an irreflexive relation without confusing the proposal with the referee's receipt?

If you submit a self-pair under this board's irreflexive relation rule, Behemoth will set fire to your coat. Equality itself may remain reflexive. If you assert that MSS-2 is $S(2)$, Koroviev will laugh until dawn. And if you replace an exact capstone calculation with an unchecked approximation, Azazello will send you back to your arithmetic."*

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3. The Retinue and the Delegation

Around the salon, the rest of the tribunal stood at their assigned posts:

- **Koroviev (Fagot)** was balanced on a rickety three-legged stool, wearing dirty checkered trousers that ended six inches above his boots, an unbuttoned vest, and a cracked pince-nez attached to his ear by dirty thread. He giggled shrilly: > *"Oh, Messire! Marvel at this magnificent mind! Look at his pockets—crammed full of 'intuitive prompts'! He thinks that if a language model writes 'trust me, this relation holds', that constitutes a proof! He doesn't know that an atlas entry is an obligation, not an admitted derivation! Oh, the tears of our educational academies!"*
- **Behemoth**, a coal-black tomcat the size of a six-month bear cub, was sprawled across the green baize chessboard on the dining table. With one enormous claw, he speared a pickled mushroom from a crystal jar; with the other, he balanced an ignited brass primus stove, its blue flame hissing ominously: > *"I am the grand master of combinatorial discipline,"* Behemoth purred, licking brine from his whiskers. *"If this candidate draws a loop connecting a mark back to itself, I shall take his knight, swallow his pawn, and pour hot kerosene down his neck."*
- **Azazello** stood motionless against the black oak wainscoting. Short, broad, in a bowler hat, with a yellow fang protruding past his lip and a wall-eyed squint fixed coldly on the candidate: > *"One hand-waving step,"* Azazello rasped, tapping a steel crowbar against the sill. *"One smuggled approximation, and you take the express route down to the tram tracks."*
- **Abadonna** sat completely immobile in the shadowy corner, wearing thick, opaque black lenses that completely hid his eyes. No one looked directly at Abadonna; everyone knew that if Abadonna removed his glasses and cast his gaze upon a manuscript, every unverified lemma and every empty claim would turn to fine grey ash.

Watching from the open door to the kitchen was the **Russian-American Delegation**:

- **Jeremy**, the 48-year-old Michigan Yooper solver from Sault Ste. Marie, leaning against the doorframe with rolled-up sleeves and a pencil behind his ear: > *"Listen to Dau, kid. Stop philosophizing about the cosmos. Declare the board. State which two marks are admitted. Write the checking*

rule. For this first finite board, give the referee a rule that another person can replay. State what passed and what you did not check."

- **Liudmila (Luda)** set down a tray with a steaming samovar, a glass in a silver holder, and a bowl of strawberry jam: > *"Stop trembling, young man. Drink your tea. Don't let Dau terrify you into silence. Keep your thoughts orderly: first the ground presentation, then an admitted mark, then a witnessed distinction. Keep the question small enough to check, and say when it remains unresolved."*
- **Polina**, 13 years old, leaned against Jeremy's shoulder with a bright, mocking grin: > *"Dad, five bucks says he writes `a = b` before he even finishes page one. Look at him, he's practically sweating through his socks. Bless his heart."*
- **Yaroslava**, 20, held a mechanical stopwatch and an aluminum clipboard: > *"Timer is running. Four rooms for the core ideas, then four optional final challenges. Keep the key closed until you have attempted them. Begemot, stop waving that primus near the paper."*

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4. The Seven Mortal Sins of the Paper 000 Examinee

Landau stepped to the slate blackboard by the window and, with three savage screeching strokes of his railway chalk, engraved the **Seven Mortal Sins**:

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================================================================================
           THE SEVEN MORTAL SINS OF THE PAPER 000 EXAMINEE
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1. THE CONFLATION OF S(n) AND MSS-N:
   Asserting that a Genesis derivation stage S(n) is identical to an atlas
   entry MSS-N. S(n) is an internal derivation governed by unresolved
   obstructions; MSS-N is an atlas of representation obligations. MSS-2 != S(2).
   Penalty: Koroviev laughs continuously into the examinee's ear for 15 minutes.

2. THE PROPOSER-CHECKER COLLUSION:
   Assuming that proposing a relation implies that the relation holds. A proposal
   packet is untrusted text; only an independent referee receipt confers status.
   Penalty: Hella confiscates the candidate's pen.

3. THE SELF-LOOP BLASPHEMY:
   Writing a relation where a mark relates to itself (x ~ x). Irreflexivity is
   declared for this finite teaching board; reflexive equality is not forbidden.
   Penalty: Behemoth overturns the primus stove.

4. THE UNCHECKED APPROXIMATION:
   Replacing the exact integer/fraction calculations in the capstones with
   rounded values and presenting them as exact certificates.
   Penalty: Automatic score = -infinity.
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5. THE UNREGISTERED SMUGGLED MARK:
   Introducing a third mark 'c' or an unadmitted symbol during a repair of
   carrier {a, b} without a new formal declaration.
   Penalty: Azazello delivers a physical reminder.

6. THE OMNISCIENT CLOSURE TRAP:
   Claiming that because the finite separation task closes at S(4), all possible
   future obstructions in the universe have been solved.
   Penalty: Abadonna lowers his black glasses by 1 millimeter.

7. THE ENVELOPE AND CONTENTS CONFUSION:
   Mistaking a balanced display or a valid JSON envelope for a checked claim.
   The bracket alphabet governs this display only. Paper 000 uses JSON packets.
   Penalty: Summary expulsion via the Patriarch's Ponds tram line.
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5. The Bracket Display for This Chalkboard Round

Landau dusted his palms on his trousers and tapped the chalk on the table:

> *"Here is your slate. Here is your chalk. You are allowed exactly 38 characters:"*

$$\Sigma_{\text{Landau}} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, b, \ldots, z, <, >\}$$

- **Ground:** in this exercise, `<>` is one opaque presentation with zero admitted carrier marks. Another register requires its own explicit reading.
- **Nesting:** Compound operations and relations are enclosed strictly by matched angle brackets: `<relation<<arg1><arg2>>>`
- **Atoms:** Lowercase alphanumeric identifiers matching `^[0-9a-z]+$`.
- **Display delimiters:** This bracket exercise uses no commas or curly braces. JSON, Python, prose, and mathematical explanations keep their own grammars.

Woland leaned forward, his green eye flaring with ancient, wicked delight:

> *"Four rooms, candidate.
> First declare the board. Then distinguish and type. Then bind and replay. Finally repair and audit.
> At the end await four optional challenges: every tied separator minimum, an exact cycle certificate, declared cells and their boundaries, and the tribunal of a substituted packet.
> The challenges have finite solutions. Your task is to earn them, and to keep their limits visible.

Dau, hand him the examination paper."*